Frenet–Serret Formulas

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of frenet–serret formulas, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

DG-01 / Differential Geometry / Report v1

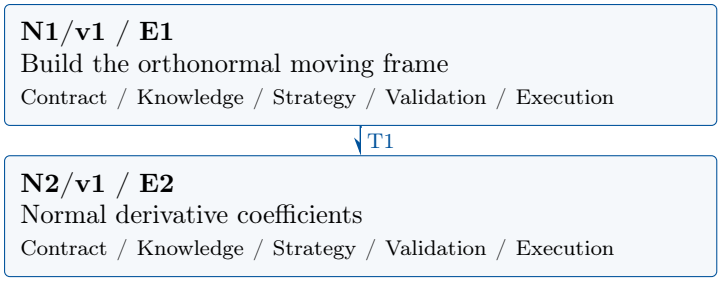
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a C^3 unit-speed curve $\gamma : I \to \mathbb{R}^3$ with curvature $\kappa = \|\gamma''\| > 0$, define $T = \gamma'$, $N = T'/\kappa$, $B = T \times N$, $\tau = -B' \cdot N$. Prove $T' = \kappa N$, $N' = -\kappa T + \tau B$, and $B' = -\tau N$.*

A. Foundations and permitted imports

Use a C^3 unit-speed curve γ with $\kappa = \|\gamma''\| > 0$. Differentiation of inner products and cross products is foundational. T,N,B form the positively oriented frame with $B = T \times N$; torsion is defined as $\tau = -B' \cdot N$. No Frenet–Serret formulas are imported.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. An orthonormal frame and $T' = \kappa N$.

Receiving activity. N2: Normal derivative coefficients

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Orthonormal frame and first formula; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Build the orthonormal moving frame

Object and role. Differentiate $T \cdot T = 1$ before normalizing.

Inputs. Unit speed; $\kappa > 0$; foundational imports stated in §1

C — Contract

Own E1/v1: An orthonormal frame and $T' = \kappa N$.

K — Knowledge

Retain Unit speed; $\kappa > 0$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Differentiate $T \cdot T = 1$ before normalizing.

V — Validation

If $\kappa = 0$ the definition of N divides by zero.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Let $T = \gamma'$.

N1.2 Unit speed gives $T \cdot T = 1$ and hence $T' \cdot T = 0$.

N1.3 Define $N = T'/\kappa$; it has unit length and is perpendicular to T.

N1.4 Then $B = T \times N$ is perpendicular to both, of unit length, and the ordered frame (T,N,B) is positive.

N1.5 By definition $T' = \kappa N$.

N1.6 C^3 regularity and $\kappa > 0$ ensure N and B are differentiable.

Export E1/v1. An orthonormal frame and $T' = \kappa N$.

Acceptance gate. If $\kappa = 0$ the definition of N divides by zero. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: If $\kappa = 0$ the definition of N divides by zero. Recheck dependent nodes N2.

B. N2/v1: Normal derivative coefficients

Object and role. Differentiate orthogonality identities to determine components.

Inputs. Orthonormal frame and first formula; prior exports E1/v1

C — Contract

Own E2/v1: $N' = -\kappa T + \tau B$ and $B' = -\tau N$.

K — Knowledge

Retain Orthonormal frame and first formula; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Differentiate orthogonality identities to determine components.

V — Validation

The torsion sign depends on the fixed $B = T \times N$ convention.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 From $N \cdot N = 1$ obtain $N' \cdot N = 0$.

N2.2 Differentiating $N \cdot T = 0$ gives $N' \cdot T = -N \cdot T' = -\kappa$.

N2.3 From $B \cdot B = 1$ and $B \cdot T = 0$ obtain $B' \cdot B = 0$ and $B' \cdot T = -B \cdot T' = 0$.

N2.4 Thus B' is a multiple of N.

N2.5 Its coefficient is $-\tau$ by the torsion definition, so $B' = -\tau N$.

N2.6 Differentiating $B \cdot N = 0$ gives $N' \cdot B = -B' \cdot N = \tau$; expanding in the frame yields $N' = -\kappa T + \tau B$.

Export E2/v1. $N' = -\kappa T + \tau B$ and $B' = -\tau N$.

Acceptance gate. The torsion sign depends on the fixed $B = T \times N$ convention. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The torsion sign depends on the fixed $B = T \times N$ convention. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Build the orthonormal moving frame

1. Let $T = \gamma'$.
2. Unit speed gives $T \cdot T = 1$ and hence $T' \cdot T = 0$.
3. Define $N = T'/\kappa$; it has unit length and is perpendicular to T.
4. Then $B = T \times N$ is perpendicular to both, of unit length, and the ordered frame (T,N,B) is positive.
5. By definition $T' = \kappa N$.
6. C^3 regularity and $\kappa > 0$ ensure N and B are differentiable.

B. §4.N2 — Normal derivative coefficients

1. From $N \cdot N = 1$ obtain $N' \cdot N = 0$.
2. Differentiating $N \cdot T = 0$ gives $N' \cdot T = -N \cdot T' = -\kappa$.
3. From $B \cdot B = 1$ and $B \cdot T = 0$ obtain $B' \cdot B = 0$ and $B' \cdot T = -B \cdot T' = 0$.
4. Thus B' is a multiple of N.
5. Its coefficient is $-\tau$ by the torsion definition, so $B' = -\tau N$.
6. Differentiating $B \cdot N = 0$ gives $N' \cdot B = -B' \cdot N = \tau$; expanding in the frame yields $N' = -\kappa T + \tau B$.

□

V. WORKED ORCHESTRATION

For the helix $\gamma(s) = (a\cos(s/c), a\sin(s/c), b\,s/c)$, where $a > 0, c = \sqrt{a^2 + b^2}$, speed is 1. Calculation yields $\kappa = a/c^2$ and $\tau = b/c^2$. For $b = 0$ the circle has $\tau = 0$ and the binormal is constant.

A. Boundary and hypothesis audit

A straight unit-speed line has $\kappa = 0$. It has many possible normal directions, so this Frenet frame is undefined there. Validation blocks division by κ; an alternative frame such as a parallel/Bishop frame would be a separately specified construction.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A straight unit-speed line has $\kappa = 0$. It has many possible normal directions, so this Frenet frame is undefined there. Validation blocks division by κ; an alternative frame such as a parallel/Bishop frame would be a separately specified construction.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.